

HBT57C V1.0 Integrated Closed-Loop Motor

Product Overview

Description

The HBT57C is a new type of integrated closed-loop motor developed by our company, based on many years of experience in low-voltage servo systems. This product utilizes the latest DSP digital processing chip and advanced variable current and variable frequency control algorithms, providing equipment manufacturers with a cost-effective hybrid servo drive solution. The HBT57C features a compact structure, small size, space savings, and reduced electromagnetic interference between wires. It adopts improved vibration and low-heat technologies to effectively solve issues such as heating, vibration, and noise in motors and drivers, resulting in lower heat output and environmental friendliness.

Features

- Operating voltage: DC input voltage 24VDC to 55VDC (recommended: 36VDC)
- Maximum continuous output current: 4A; peak: 5A
- Accepts both differential and single-ended pulse/direction commands
- FOC and SVPWM closed-loop control
- Variable current and frequency for reduced heating
- Adjustable pulses per revolution via software or DIP switches
- Overvoltage, undervoltage, overcurrent, and following error protection
- Selectable single/dual pulse mode and pulse edge triggering (via PC)
- Max pulse frequency: 500KHz (default: 200KHz)
- Pulse, direction, enable input voltage: 5V-24V (no resistor needed)
- Serial debugging function (special cable)
- Performance: Smooth speed, minimal overshoot, small tracking error, low heat

Applications

Suitable for small/medium automation equipment: screwdrivers, wire strippers, winding machines, terminal machines, laser machines, printers, engraving machines, electronic processing, pick-and-place, CNC, packaging, robots. Especially where low noise and high speed are needed.

LED Status Indicator

The green LED serves as the power indicator. When the driver is connected to the power supply, the LED remains steadily lit; when the driver is disconnected from the power supply, the LED turns off.

The red LED acts as a fault indicator. In the event of a fault, the indicator will flash cyclically with a 5-second period. Once the fault has been cleared by the user, the red LED remains steadily lit. The flashing frequency of the red LED is 2 Hz, with the LED on for 200 ms and off for 300 ms. The number of red LED flashes within 5 seconds indicates different types of fault information, as detailed in the table below:

No.	Number of Flashes	Red LED Flash Pattern	Fault Description
1	1		Overcurrent fault ($I \geq 6A$)
2	2		Overvoltage fault ($V \geq 60V$)
3	7		Excessive tracking error fault

Note:

When a fault occurs, the driver will stop output and display the corresponding fault code. The user must power off and then restart the driver before the fault can be cleared.

Electrical, Mechanical, and Environmental Specifications

Electrical Specifications

Parameter	Min	Typical	Max	Unit
Continuous Output Current	0.5	–	4	A
Power Supply Voltage (DC)	24	36	55	Vdc
Logic Input Current	10	10	16	mA
Logic Input Voltage	4.5	5	24	Vdc
Pulse Frequency	0	200	500	kHz
Pulse High Width	1.5	–	–	μs
Position Accuracy	–	±1	–	Pulse
Speed Accuracy	–	±2	–	rpm
Max Acceleration (no load)	–	100	–	rpm/ms
Overvoltage Protection	55	60	65	Vdc
Insulation Resistance	100	–	–	MΩ

Environmental Parameters

Cooling: Natural or forced air cooling

Location: Avoid heat, dust, oil, corrosives, humidity, vibration, flammables, conductive dust

Working Temp: -5°C ~ +45°C

Humidity: 40~90% RH

Vibration: 10~55Hz / 0.15mm

Storage Temp: -20°C ~ +65°C

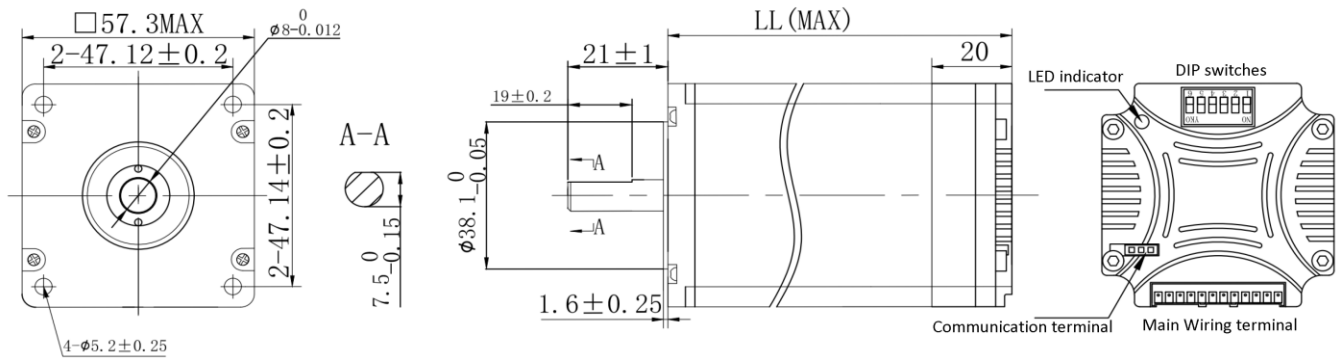
Altitude: ≤1000m

Weight: See below

Mechanical Dimensions

Model	Body Length (mm)	Shaft Length (mm)	Shaft Diameter (mm)	Phase Current (A)	Phase Resistance (Ω)	Holding Torque (N.m)	Inertia (g.cm ²)	Weight (g)
HBT5756C	76	21	8	3.0	0.93	1.5	300	750
HBT5776C	96	21	8	3.5	0.40	2.0	450	1100
HBT57100C	120	21	8	4.0	0.50	2.8	680	1560
HBT57112C	132	21	8	4.5	0.78	3.2	760	1780

Mechanical Drawing:



Note: For demanding low-speed vibration or high-speed performance, contact us for algorithm optimization.

Heat Dissipation Precautions

- Environment temp: -5°C ~ 45°C
- Motor temp: ≤65°C
- If needed, install fan for forced cooling.

The reliable operating ambient temperature for the integrated closed-loop motor is typically between -5°C and 45°C. The motor's temperature during operation should not exceed 65°C. If necessary, a fan should be installed at the rear cover of the motor to provide forced cooling, ensuring the motor operates within the specified reliable temperature range.

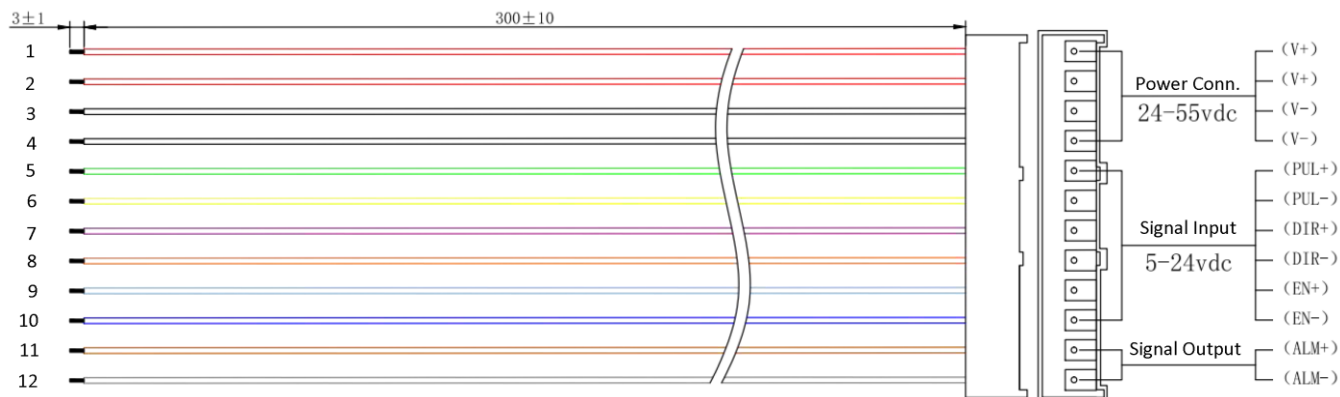
Driver Interface and Wiring Instructions

Interface Description

Control Port: 12-pin white terminal

Pin Assignments:

PIN No.	Signal	Color	Function	Description
1-2	+VDC	Red	Power input positive, input voltage 24–55VDC	(Note: Pay attention to power polarity, do not reverse!!!)
3-4	GND	Black	Power input ground	
5	PU+	Green	Signal positive input terminal	Compatible with 4.5–24V level signals
6	PU-	Yellow	Signal negative input terminal	
7	DR+	Purple	Signal positive input terminal	
8	DR-	Orange	Signal negative input terminal	
9	EN+	White	Signal positive input terminal	
10	EN-	Blue	Signal negative input terminal	
11	AL+	Brown	Alarm signal positive output terminal	OC open-collector output, max pull-up voltage 24V, max output current 100mA
12	AL-	Gray	Alarm signal negative output terminal	



RS232 Serial Communication Interface

You can connect to a PC via a serial converter (sold separately) and the dedicated serial cable (do not hot-plug while powered). The PC software allows you to configure driver functions and parameters, such as microstep settings, current values, and resonance suppression as needed by the user.

Pin Number	Symbol	Name	Description
1	+5V	5V Power Output	For external STU use only
2	TXD	RS232 Transmit	
3	RXD	RS232 Receive	
4	GND	5V Power Ground	0V

Note: The serial cable must be the dedicated cable provided. Depending on user circumstances, it may be included. Please confirm before use to avoid damage.

Control Signal Interface Circuit

The HBT57C driver’s signal input interface can accept differential signal input, single-ended common-anode signal input, and single-ended common-cathode signal input. It is equipped with a high-speed optocoupler for isolation; the output is an open-collector (OC) transistor. The interface connection methods are as follows:

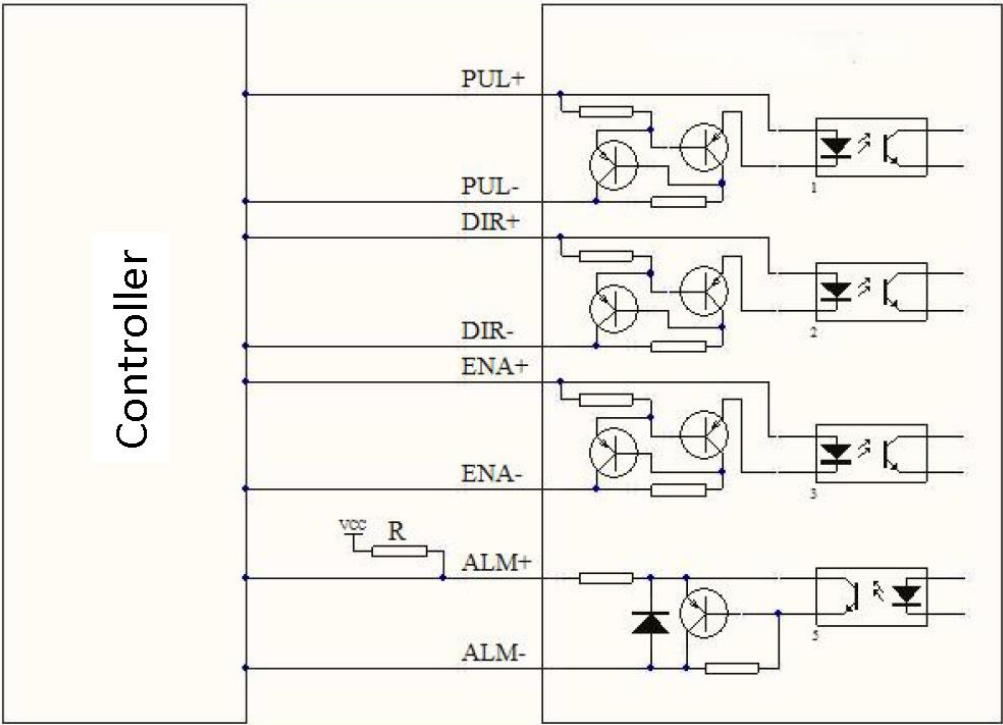


Figure 1 Differential Signal Input Wiring

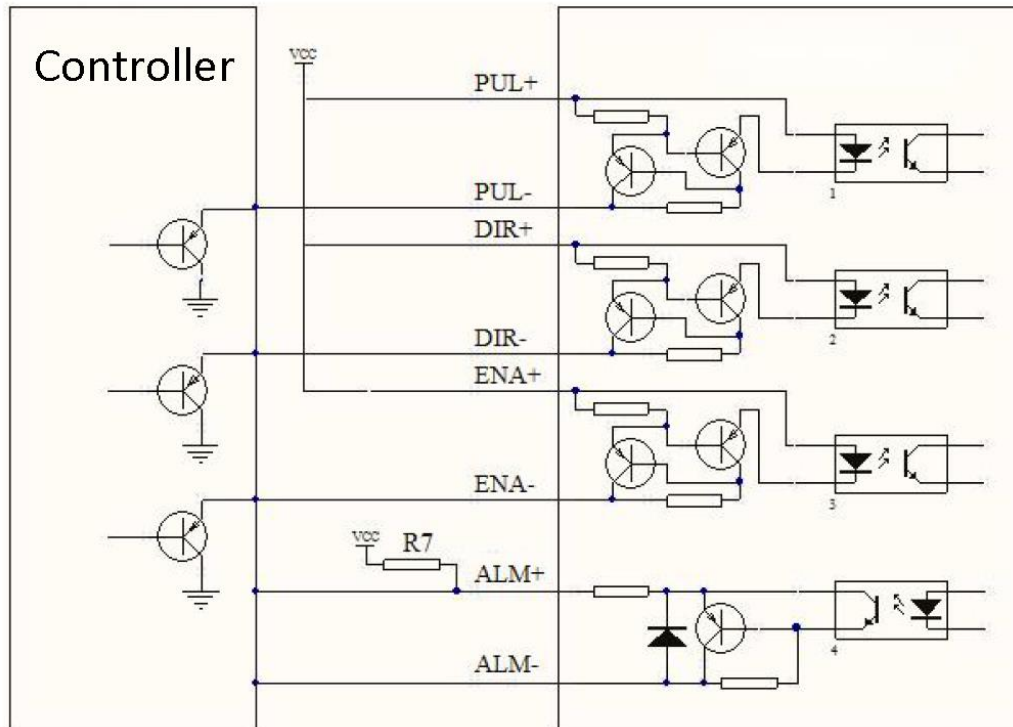


Figure 2 Single-Ended Common-Anode Signal Input Wiring

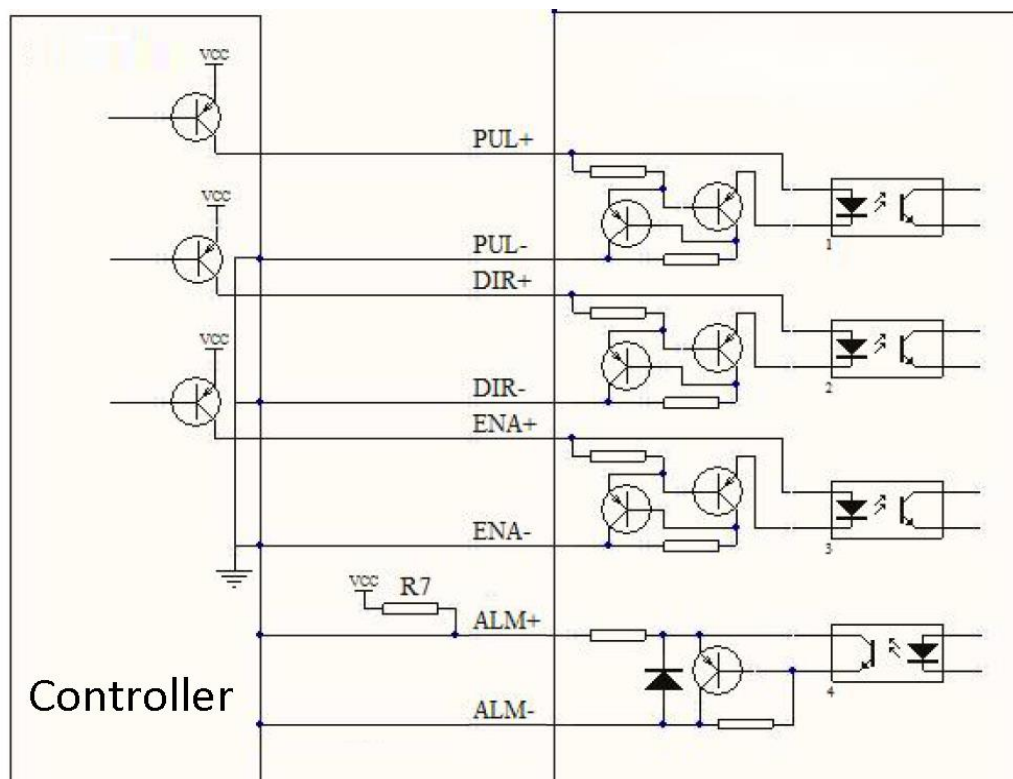


Figure 3 Single-Ended Common-Cathode Signal Input Wiring

Note:

The signal input interface voltage range in the diagram is 5V. Regardless of whether it is single-ended or differential input, there is no need to connect a current-limiting resistor in series.

For the output interface, the maximum external pull-up voltage is 28Vdc and the maximum output current is 100mA. Select an appropriate pull-up resistor according to the external pull-up voltage. As a basic reference:

- If the external pull-up voltage is 24Vdc, use a 2K pull-up resistor.
- If the external pull-up voltage is 12Vdc, use a 1K pull-up resistor.
- If driving a relay or motor brake coil, please consult our application engineer.

Control Signal Timing Diagram

To avoid incorrect actions and deviations, PUL-, DIR-, and ENA- must meet certain requirements, as illustrated below:

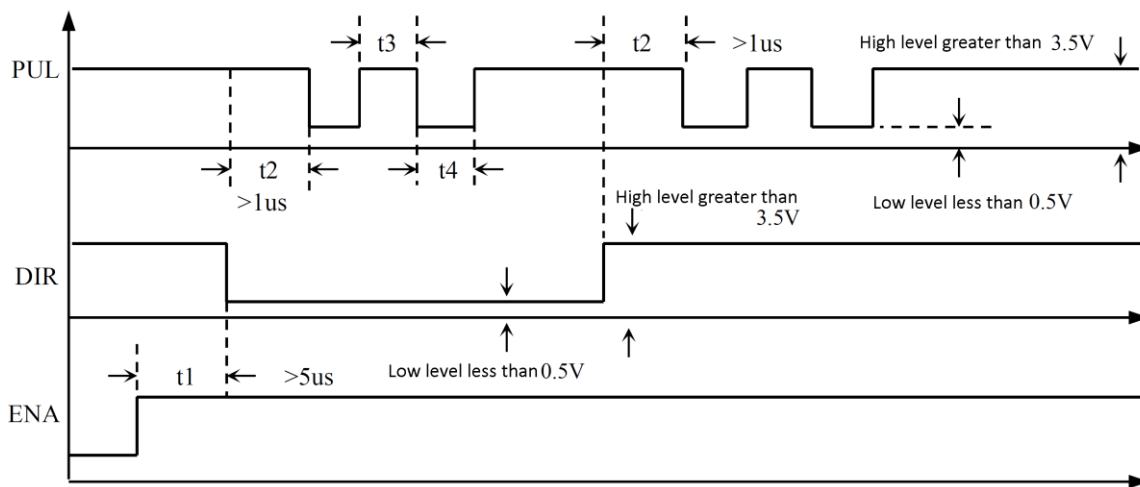


Figure 5 Control Signal Timing Diagram

Notes:

1. t_1 : ENA (Enable signal) should be set high at least $5 \mu s$ before DIR. In general, it is recommended to leave it floating.
2. t_2 : DIR should be set high or low at least $1 \mu s$ before the falling edge of PUL.
3. t_3 : The pulse width should be no less than $1.5 \mu s$.
4. t_4 : The low-level width should be no less than $1.5 \mu s$.

Control Signal Mode Settings

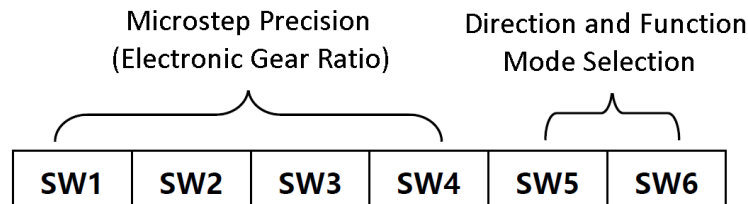
- **Pulse Trigger Edge Selection:**
Use PC software to set the effective pulse trigger edge as either rising or falling edge.
- **Single/Dual Pulse Selection:**
Use PC software to set whether single pulse or dual pulse is effective.
- **Direction Selection:**
Use PC software to set the initial running direction of the motor.

Wiring Requirements

1. To prevent the driver from being affected by interference, it is recommended to use shielded cables for control signals, with the shielding layer grounded. Except for special requirements, the shielding layer of the control signal cable should be grounded at one end only: at the controller (upper computer) side, and left unconnected at the driver side. In one machine, only one point is allowed for grounding. If it is not a real ground, severe interference may occur, so in such cases, the shield should not be connected. If possible, thermal grounding technology is most effective for shielding.
2. Pulse and direction signal wires must not be bundled or routed together with high-voltage power wires. It is best to separate them by at least 10 cm. Otherwise, power noise may interfere with the pulse and direction signal wires, causing positioning errors or system instability.
3. If one power supply is used for multiple drivers, parallel connection should be used at the power source. Do not connect from one driver to another in a daisy chain.
4. It is strictly forbidden to plug or unplug driver terminals while powered. When powered, a large current may still be flowing through the coils, and unplugging will cause a large instantaneous induced voltage, which can burn out the driver.
5. It is strictly forbidden to connect wires with tinned ends to the terminals, as it may result in increased contact resistance and overheating, damaging the terminal.
6. The exposed part of the wire must not protrude outside the terminal to prevent accidental short circuits and damage to the driver.

DIP Switch Settings

The HBT57C digital integrated closed-loop motor uses a 6-position DIP switch to set microstep precision (electronic gear ratio), initial rotation direction, and function mode selection. Details are as follows:



Pulse/rev	SW1	SW2	SW3	SW4
Default	ON	ON	ON	ON
800	OFF	ON	ON	ON
1600	ON	OFF	ON	ON
3200	OFF	OFF	ON	ON
6400	ON	ON	OFF	ON
12800	OFF	ON	OFF	ON
25600	ON	OFF	OFF	ON
51200	OFF	OFF	OFF	ON
1000	ON	ON	ON	OFF
2000	OFF	ON	ON	OFF
4000	ON	OFF	ON	OFF
5000	OFF	OFF	ON	OFF
8000	ON	ON	OFF	OFF
10000	OFF	ON	OFF	OFF
3600	ON	OFF	OFF	OFF
7200	OFF	OFF	OFF	OFF

Microstep Precision (Electronic Gear Ratio):

When S1, S2, S3, and S4 are all set to ON, the driver uses the internal default microstep setting, with the initial value of 400 Pulse/rev. You can also set the electronic gear ratio via the tuning software.

SW5 sets the motor direction: when ON, the motor rotates clockwise (CW); when OFF, it rotates counterclockwise (CCW).

SW6 selects the function mode: when OFF, the driver is in field-oriented control mode (FOC); when ON, the driver is in point-to-point (positioning) mode (PM), which provides better start/stop performance.

Protection Functions

- **Overvoltage Protection**

When the input voltage of the HBT57C exceeds 55Vdc, the driver will stop operating. You must clear the fault and power cycle to reset.

- **Undervoltage Protection**

When the input voltage of the HBT57C falls below 18Vdc, the driver will stop operating. You must clear the fault and power cycle to reset.

- **Overcurrent Protection**

When an overcurrent occurs, the HBT57C driver will stop operating. You must clear the fault and power cycle to reset.

- **Tracking Error Exceeded**

When a tracking error exceeds the limit, the HBT57C driver will stop operating. You must clear the fault and power cycle to reset.

Attention:

Since the driver does not have power supply reverse polarity protection, please double-check the positive and negative wiring of the power supply before powering on. Reversing the polarity will cause the fuse in the driver to blow!

Common Issues

Application Problems and Solutions

Issue	Possible Cause	Solution
Motor not running	Power LED off	Check power circuit
	Shaft engaged but not rotating	Increase signal current to 7–16mA
	Incorrect subdivision	Select correct subdivision
	Driver protection active	Power cycle
	Enable signal low	Pull enable high or disconnect
	No response to control signal	Power not on
	Motor wire open	Check and reconnect
	Voltage too high/low	Check power supply
	Motor or driver damaged	Replace motor or driver
Position error	Signal interference	Eliminate interference
	Shield not grounded	Ground shield
	Motor wire open	Check and reconnect
	Incorrect subdivision	Set correct subdivision
Stalling (accel.)	Acceleration too short	Increase acceleration time
	Motor torque too low	Select higher torque motor
	Low voltage	Increase voltage

Frequently Asked Questions and Answers about the Driver

1) What are the advantages of a microstepping servo driver?

- By reducing the step angle for each step, the uniformity of each step is improved, thus enhancing control accuracy.
- It can greatly reduce motor vibration. Low-frequency resonance is an inherent characteristic of stepper motors, and using microstepping is the best way to eliminate it.
- It can effectively reduce torque ripple and increase output torque.
- All of the above advantages are widely recognized by users and provide tangible benefits, so it is recommended to use a microstepping driver.

2) Why does my motor only rotate in one direction?

- The direction signal may be too weak, the wiring polarity may be incorrect, or the signal voltage may be too high, burning out the direction current-limiting resistor.
- The pulse mode is not matched; if the signal is pulse/direction, the driver must be set to this mode.

